

Review and Reconstruction of Autonomous Vehicle Governance Frameworks from a Comparative Law Perspective

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Abstract. The deep integration of autonomous driving technology and the automotive industry is reshaping global transportation and challenging traditional vehicle legal systems. Major countries and regions have accelerated legislation, but there are disparities in regulatory philosophies, technical standards and liability allocation, causing cross border legal conflicts and regulatory gaps. This paper uses a comparative law approach to analyze six typical governance models (the United Nations, the European Union, Germany, the United Kingdom, the United States and Japan), extracting common experiences and core divergences in safety assessment, access management and liability configuration. It then examines structural problems in China's autonomous driving governance, such as the absence of superior legislation, fragmented local rules, imperfect liability systems and insufficient international coordination. Finally, it proposes a governance reconstruction scheme based on "unified legislation, classified regulation and international coordination". It suggests enacting an Autonomous Vehicles Law, establishing a two - tier governance structure, improving the safety assessment system, and participating in international rule - making at the UN level to guarantee the large scale application of autonomous driving technology in China.

1. Introduction

As a key application of artificial intelligence, autonomous driving is driving a historic shift in transportation from human-driven to system-driven vehicles. However, a clear tension exists between the rapid iteration of technology and the relative stability of legal systems. Traditional road traffic safety laws are based on the fundamental assumption that human drivers always control vehicles. As driving tasks gradually shift to algorithmic systems, existing systems for vehicle access standards, accident liability determination and data security management face applicability dilemmas^[1].

To address this challenge, major global economies have launched legislative processes, forming distinctive governance models. The EU promotes regional unified standards under the UN

framework; the US has a decentralized governance structure between federal and state levels; Germany and Japan balance strict technical regulations with gradual opening-up. While this diversified institutional exploration provides trial-and-error space for countries, it also causes problems such as fragmented technical standards, high cross-border operation compliance costs and conflicting accident liability jurisdictions. Importantly, the UN global technical regulation for Automated Driving Systems (ADS) will be completed for review in 2026, placing the international rule system at a critical formative stage. How to effectively express China's institutional demands in this process has become an urgent issue.

Through comparative analysis of major global autonomous driving governance paradigms, this paper summarizes institutional experiences for reference and proposes systematic governance reconstruction suggestions combined with China's industrial development and legal construction realities. The research aims to provide theoretical references for revising China's Road Traffic Safety Law and enacting dedicated autonomous driving legislation, promoting the establishment of an autonomous driving legal system that conforms to China's national conditions and aligns with international rules^[2].

2. Comparative Analysis of International Autonomous Driving Governance Frameworks

Current global autonomous driving governance presents a multi-level pattern of "UN coordination, regional promotion and national implementation". The UN works to develop globally unified technical regulations; regional organizations such as the EU build supranational access systems; and countries retain autonomy in traffic behavior regulation and liability allocation. This nested governance structure reflects both the objective requirement of internationalized technical standards and the differences in national legal traditions and industrial interests.

2.1 UN Framework: International Convention Revision and Global Technical Regulation Formulation

The United Nations Economic Commission for Europe (UNECE) is the core platform for international autonomous driving governance, focusing on two dimensions: revising the Vienna Convention on Road Traffic to provide an international legal basis for autonomous vehicles on public roads, and formulating binding vehicle technical regulations to unify national safety standards.

The 2014 amendment to the 1968 Vienna Convention on Road Traffic first clarified that autonomous driving systems can take over driving duties under specific conditions. It allows autonomous systems to operate on public roads as long as they comply with UN vehicle type approval regulations and human drivers can intervene or shut down the system at any time. However, its application scope is limited as major countries such as China, the US and Japan are not parties to the convention^[3].

In terms of technical regulations, the UNECE Working Party on Automated/Connected Vehicles (GRVA) has established a relatively complete regulatory system. UN R155 (Cybersecurity), UN R156 (Software Updates) and UN R157 (Automated Lane Keeping Systems) constitute the core basis for L3 autonomous driving type approval. UN R157, the world's first L3 autonomous driving regulation, has undergone two important revisions: the 2020 initial version limited the maximum speed to 60 km/h and prohibited lane changes, while the 2022 revision raised the speed limit to 130 km/h and added lane change functionality, reflecting a regulatory approach of gradually relaxing restrictions as technology matures.

More prospectively, GRVA is integrating the Framework for Regulatory Assessment and Verification of ADS Safety (FRAV) and the New Assessment/Test Method for Automated Driving (NATM) into a universally binding UN global technical regulation, scheduled for submission to WP.29 for adoption in June 2026. The core innovation of this regulatory system is the establishment of a "multi-pillar assessment method", which comprehensively verifies the safety of autonomous driving systems through simulation testing, closed-field testing, real-road testing and audit evaluation. This marks a major shift in vehicle safety regulation from traditional "compliance testing" to "safety argumentation".

2.2 EU: Supranational Unified Type Approval Framework

The EU adopts a dual-track strategy of "relying on the UN framework plus supplementary independent legislation", building a complete access specification system covering L3 to L4 autonomous driving based on Regulation (EU) 2018/858 (Type Approval Framework Regulation) and Regulation (EU) 2019/2144 (General Safety Regulation).

For L3 autonomous driving, the EU directly incorporates UN regulations such as UN R155 and UN R157 into the EU type approval system through Delegated Regulation (EU) 2022/1398, achieving seamless connection between international standards and regional systems and effectively avoiding rule fragmentation caused by member states formulating their own standards. For L4 autonomous driving, which has not yet been regulated by the UN, the EU first fills the gap through Implementing Regulation (EU) 2022/1426, the world's first written

regulation clarifying L4 autonomous driving type approval procedures and technical requirements. This regulation currently implements "small-batch" restrictions (no more than 1,500 units per model per year) and plans to lift the quantity limit in 2026 and align with the UN ADS general regulation.

A notable feature of the EU governance framework is adherence to the "technology neutrality" principle: regulations only set safety performance targets without mandating specific technical implementation paths, leaving room for competition between different technical routes. This concept is consistent with the governance logic of the EU Artificial Intelligence Act for "high-risk AI systems", reflecting a modern regulatory approach based on risk and performance.

2.3 Germany: Two-Tier Access and Liability System Reconstruction

As a traditional automotive power, Germany leads Europe in autonomous driving legislation. The 8th amendment to the Road Traffic Act in 2017 first allowed highly or fully autonomous driving systems to operate on public roads, establishing the basic framework that "systems can take over driving tasks but human drivers must remain ready to take over". The Autonomous Driving Act introduced in 2021 went further, opening legal channels for the commercial operation of L4 autonomous driving in designated operating areas, including urban shuttles, trunk logistics and demand-responsive travel services.

The core innovation of Germany's institutional design is the establishment of a "three-step approval" procedure: first, the Federal Motor Transport Authority (KBA) conducts vehicle type approval in accordance with EU regulations, reviewing technical requirements such as safety concepts and multi-pillar verification results; second, state-level authorities approve designated operating areas, evaluating factors such as infrastructure adaptability and traffic environment impact; third, complete motor vehicle registration and license issuance. This progressive structure of "access approval + regional approval + license registration" achieves institutional separation between vehicle technical safety and operating environment safety, providing an important reference for other countries^[4].

In terms of liability allocation, while retaining the basic framework of no-fault liability for vehicle owners, German law added a technical supervisor system through the Autonomous Driving Act, requiring L4 vehicle operations to be remotely monitored by natural persons who can intervene at any time. This "human-machine dual-track" liability distribution model solves the problem of absent liability subjects in fully driverless scenarios and alleviates the attribution dilemma of algorithmic decisions by retaining human intervention nodes.

2.4 UK: Insurance-Led Liability Allocation Model

The UK Autonomous Vehicles Act 2024 represents a unique exploration of common law systems in autonomous driving governance, with its core logic being dispersing technical risks through insurance mechanisms while retaining the right to recover from the ultimately liable party.

The act stipulates that when an accident occurs while an autonomous vehicle is in "autonomous mode", the insurance company first bears compensation liability to the victim; after paying compensation, the insurance company has the right to exercise subrogation against the truly liable party (usually the vehicle manufacturer or autonomous driving system developer). This "compensate first, recover later" mechanism integrates product liability logic into traffic accident compensation systems, allowing victims to obtain timely relief without proving the manufacturer's fault, while manufacturers also achieve decentralized liability risks through the insurance market. In addition, the act created the concept of "authorized autonomous entities", clarifying that criminal liability during autonomous system control rests with enterprises rather than drivers, and established an independent accident investigation mechanism.

The advantage of the UK model is that it responds to the attribution problem with relatively moderate institutional adjustments, neither radically granting legal personality to autonomous driving systems nor rigidly maintaining the traditional driver liability framework. However, its limitations are also obvious: it relies heavily on mature insurance markets and actuarial systems, and cross-border recovery may face practical obstacles such as jurisdictional conflicts.

2.5 US: Fragmented Governance under Federalism

US autonomous driving governance presents a distinct "federal-state" two-tier structure: at the federal level, the National Highway Traffic Safety Administration (NHTSA) is responsible for formulating vehicle safety standards and product access rules; at the state level, authorities are responsible for driver licensing, vehicle registration, traffic law enforcement and liability insurance supervision. This separation of powers under the constitutional framework has led to a governance pattern of "lagging federal rules, active state legislation and lack of mutual recognition of rules".

At the federal level, although NHTSA has issued four editions of autonomous driving policy guidelines, these guidelines are not legally binding. The "Autonomous Vehicle Safety, Transparency and Evaluation Program" (AV STEP) proposed in December 2024 attempts to build a voluntary regulatory framework, where participating enterprises can obtain flexibility

to exempt from current safety standards (such as requirements for manual control devices like steering wheels and brake pedals) by submitting safety assurance plans and safety incident data. The SELF DRIVE Act introduced in the House of Representatives in February 2026 aims to establish NHTSA's clear regulatory authority over autonomous driving systems, create a national autonomous driving safety data repository, and codify requirements for cybersecurity and incident response plans^[5].

At the state level, legislative activities are active but highly diverse. Nevada was the first to authorize autonomous driving road tests in 2011, while California allows companies such as Waymo to operate paid passenger services in San Francisco and Los Angeles. However, inconsistent regulations across states in testing conditions, safety officer requirements and insurance limits result in high compliance costs for cross-state operators, requiring the same autonomous driving system to adjust operating parameters for different states and severely restricting industrial scale development.

2.6 Japan: Compromise Path of Legal Revision and Guideline Guidance

Japan's autonomous driving governance adopts a dual-track model of "revising existing laws + formulating technical guidelines", with its institutional evolution closely linked to the SAE autonomous driving classification standards. For L3 autonomous driving, Japan did not enact dedicated legislation but created space for autonomous systems to legally operate within the existing legal framework by revising the Road Transport Vehicle Act and Road Traffic Act in 2020. In terms of liability allocation, the Outline for System Preparation Related to Autonomous Driving established a layered liability system: accidents are generally borne by vehicle owners; if caused by system defects, manufacturers are liable; if caused by external factors such as hacker attacks, they are resolved through government compensation mechanisms.

For L4 autonomous driving, the amendment to the Road Traffic Act that came into effect in April 2023 made breakthrough provisions: it clearly allows driverless L4 vehicles to operate on roads within specific Operational Design Domains (ODD), while setting supporting requirements such as remote monitoring obligations, speed limits and operating area approvals. Japan plans to expand the scope of application of the amendment to 40 regions nationwide by 2025, marking the transition of autonomous driving from the "testing phase" to the "normalized operation phase".

Japan's experience shows that responding to technological changes by revising existing laws rather than "starting from scratch" can effectively reduce legislative costs and institutional

coordination difficulties. However, the applicability boundary of this model is constrained by the conceptual framework and normative structure of existing laws. When technological evolution breaks through the carrying capacity of existing concepts, dedicated legislation remains irreplaceable.

2.7 Governance Consensus and Divergences

A comprehensive examination of the six governance paradigms reveals the following basic consensus: first, all adopt a phased legislative approach, gradually opening application scenarios according to technological maturity; second, all require autonomous driving systems to meet strict safety standards and use multi-pillar methods for verification; third, all recognize the limitations of traditional driver liability frameworks and actively explore liability system innovations; fourth, all incorporate cybersecurity, software updates and data recording into the core regulatory scope.

Meanwhile, significant divergences persist among countries on key issues: first, different regulatory philosophies—the EU emphasizes technology neutrality and risk classification, the US tends toward market orientation and voluntary compliance, and Germany and Japan focus on balancing strict technical specifications with limited scenario opening; second, different liability rules—Germany maintains owner no-fault liability and adds technical supervisors, the UK absorbs liability risks through insurance mechanisms, and Japan adopts a layered "owner-manufacturer-government" liability system; third, different data governance—the EU emphasizes data minimization and purpose limitation based on GDPR, the US has not yet formed a unified federal data protection framework, and China has established a security assessment mechanism for important data outbound transfers.

This institutional diversity reflects differences in national legal traditions, industrial endowments and governance philosophies, but also lays the groundwork for cross-border legal conflicts. Issues such as difficult mutual recognition of type approvals, conflicting liability jurisdictions and barriers to cross-border data flow urgently require higher-level international coordination.

3. Current Status and Problems of Autonomous Driving Governance in China

3.1 Governance Status: Local Leadership Followed by Central Action

China's autonomous driving governance presents a "bottom-up" institutional generation path. Early exploration mainly focused on road testing and demonstration applications. In 2018, the Ministry of Industry and Information Technology (MIIT), the Ministry of Public Security and the Ministry of Transport jointly issued the Administrative Specifications for Road Testing of Intelligent Connected Vehicles (Trial), which was revised to the Administrative Specifications for Road Testing and Demonstration Applications in 2021, providing unified policy guidance for local practices. As of September 2024, China has built 17 national-level testing demonstration zones and 7 internet of vehicles pilot zones, issued more than 7,700 test licenses, and accumulated over 70 million kilometers of total test mileage^[6].

In terms of product access, the four ministries jointly issued the Notice on Launching Pilot Work on Intelligent Connected Vehicle Access and Road Traffic in November 2023, establishing the first national-level access management framework for L3/L4 autonomous driving. This framework takes "enterprise capability assessment + product technology assessment" as dual thresholds, requiring enterprises to establish a full lifecycle management system covering design verification, safety assurance, safety monitoring and user notification, and verify product safety through a multi-pillar testing method combining simulation, field, road and audit evaluations. In June 2024, pilot consortia composed of 9 vehicle manufacturers and operators were approved to conduct empirical operations in 7 cities including Beijing, Shanghai and Guangzhou, marking a key leap in China's autonomous driving governance from "testing exemption" to "conditional access".

In terms of local legislation, more than ten cities including Shenzhen (2022), Shanghai Pudong (2023), Beijing (2025) and Guangzhou (2025) have issued local regulations on autonomous driving, pioneering trials in road testing, demonstration operations and commercialization pilots. These local legislatures have distinctive application scenario designs—for example, Beijing covers personal passenger travel and urban operation support, while Guangzhou focuses on hub transportation and commercial operations—accumulating valuable experience for national-level legislation.

3.2 Structural Deficiencies

Despite phased progress, China's autonomous driving governance still has deep-seated structural problems restricting large-scale technological application:

First, the contradiction between the absence of national legislation and fragmented local rules is prominent. Superior laws such as the Road Traffic Safety Law still take human drivers as the regulatory core, leaving no clear national-level provisions on core issues such as the legal status of autonomous driving systems, the legal road access qualification of L4 driverless vehicles and accident liability allocation. Although local legislatures have actively explored, they face institutional bottlenecks of "lack of authority to stipulate civil and criminal liabilities and difficulty breaking through superior law constraints". This legislative pattern of "local trials first, central lag" leads to differentiated compliance requirements for cross-regional operators and increases institutional costs for industrial development.

Second, the access standard system is imperfect and safety assessment methods need refinement. Although the current access pilot has established a multi-pillar testing framework, regulatory gaps still exist in multiple key links: there is no unified standard for simulation toolchain credibility assessment, resulting in insufficient horizontal comparability of simulation test results across enterprises; the structural requirements for safety argumentation are unclear, leading to significant differences in the completeness of argument chains and sufficiency of evidence in safety cases submitted by enterprises; the specific indicators, trigger thresholds and data reporting formats of in-use monitoring and reporting mechanisms have not been standardized, affecting the effective operation of the "monitoring-feedback-optimization" closed loop.

Third, the absence of liability rules leads to legal vacuums in accident handling. Current laws lack clear provisions on liability allocation for autonomous driving accidents: in L3 human-machine co-driving scenarios, the ownership of the "liability gap" between system takeover requests and actual driver takeover is unclear; in L4 driverless scenarios, there is no authoritative conclusion on whether the liability subject should be the vehicle owner, operation platform, manufacturer or system developer. Although current pilots require enterprises to purchase traffic accident liability insurance of no less than 5 million yuan per vehicle, insurance mechanisms can only disperse risks and cannot replace the institutional function of liability rules.

Fourth, cross-border legal coordination mechanisms are missing. China has not acceded to the UNECE 1958 Agreement, so UN technical regulations have no direct binding force on China.

Although China participates in international rule discussions through GRVA informal working groups, institutional discourse power still needs improvement. Issues such as international mutual recognition of type approvals, compliance channels for cross-border data flow and jurisdictional coordination of cross-border accidents lack systematic institutional arrangements, restricting the global layout of Chinese autonomous driving enterprises.

4. Reconstruction Path of China's Autonomous Driving Governance System

4.1 Lessons from Comparative Law Experiences

Foreign experiences provide useful references for China's governance reconstruction, but we must be alert to the "institutional trap" of simple transplantation. The value of Germany's two-tier access model lies in achieving institutional separation between vehicle technical compliance and operating environment adaptation, but its effective operation depends on clear division of powers between the federal and state governments, requiring adaptive transformation under China's unitary state structure. The advantage of the UK's "insurance first" model is resolving liability uncertainty through market mechanisms, but its application premise is highly mature insurance actuarial capabilities and reinsurance markets. China's autonomous driving insurance is still in its infancy, so comprehensive copying is inappropriate. The lesson from the US federal-state two-tier governance warns us that excessive reliance on local leadership without a unified superior law framework can easily lead to rule fragmentation and market segmentation^[7].

Therefore, the localization transformation of foreign experiences must adhere to three criteria: first, institutional adaptability—whether foreign rules are compatible with China's legal traditions, industrial endowments and governance structures; second, timeliness—whether institutional transplantation conforms to China's autonomous driving technology development stage and market scale characteristics; third, cost controllability—whether compliance costs brought by institutional changes are within the industrial affordability range.

4.2 Reconstruction Scheme

4.2.1 Enact the Autonomous Vehicles Law to Establish a Unified Legal Framework

Given that revising the Road Traffic Safety Law can only solve partial problems such as autonomous vehicle road access qualifications and cannot systematically respond to full-chain institutional demands including product access, liability allocation, data governance and

regulatory systems, it is recommended to enact a dedicated Autonomous Vehicles Law as the basic law in the autonomous driving field. The law should clarify the following core systems:

Establish the legal concept and classification standards for autonomous driving systems, replacing the cumbersome L0-L5 technical classification with a binary division of "driver assistance" and "autonomous driving" to enhance the operability of legal application;

Establish a two-tier governance structure of "product access - traffic behavior", with the State Council's industry and information technology authority responsible for product access focusing on vehicle safety performance and system reliability, and public security traffic management authorities responsible for traffic behavior regulation focusing on driving rules and accident handling;

Clarify basic liability allocation rules: in L3 human-machine co-driving mode, manufacturers bear product liability during the period when the system performs dynamic driving tasks, and drivers bear fault liability if they fail to fulfill takeover obligations; in L4 driverless mode, operating entities bear no-fault liability, and manufacturers bear product defect liability;

Establish a special chapter on autonomous driving data governance, clarifying data classification and grading standards, personal information protection rules and security assessment procedures for important data outbound transfers.

4.2.2 Build a Binary Licensing Structure of "National Unified Access - Local Scenario Approval"

Drawing on German experience and combining China's national conditions, build a binary licensing system of "national unified access + local scenario approval": at the national level, MIIT is uniformly responsible for autonomous vehicle type approval and formulates national unified safety standards and technical specifications to ensure the bottom-line requirements of product safety; at the local level, governments at or above the prefecture-level city level approve specific operating areas and application scenarios according to local traffic environments, infrastructure conditions and public safety needs. This structure not only maintains the institutional foundation of the national unified market but also retains local flexibility to adapt to local conditions, effectively resolving the tension between "unified regulation" and "differentiated demands".

4.2.3 Improve the Technical Specification System for Multi-Pillar Safety Assessment Methods

Based on the current access pilot framework, accelerate the standardization construction of multi-pillar safety assessment methods: first, formulate simulation test credibility assessment

standards, clarifying the verification process, accuracy requirements and consistency determination methods with real vehicle tests for simulation toolchains; second, issue structured guidelines for safety argumentation, standardizing the hierarchical structure of safety statements-arguments-evidence to enhance the reviewability and comparability of safety cases; third, establish a unified in-use monitoring indicator system, clarifying safety incident classification standards, data reporting formats and trigger response thresholds to provide data support for dynamic access adjustments; fourth, improve construction specifications for the Safety of the Intended Function (SOTIF) scenario library, systematically covering nominal scenarios, edge scenarios, failure scenarios and ODD boundary scenarios.

4.2.4 Actively Participate in International Rule-Making and Build Cross-Border Coordination Mechanisms

Accelerate the feasibility assessment of acceding to the UNECE 1958 Agreement to lay the institutional foundation for domestic transformation of UN technical regulations and mutual recognition of type approvals. In the short term, establish type approval mutual recognition mechanisms with major trading partners through bilateral or multilateral memoranda to reduce enterprises' overseas compliance costs. Regarding cross-border data issues, refer to the exemption situations established in the Guidelines for the Security of Outbound Data Transfers in the Automotive Sector, promote negotiations with the EU on mutual recognition of Standard Contractual Clauses (SCC), and reserve institutional channels for the international expansion of business models such as Robotaxi. Meanwhile, deeply participate in the formulation process of the UN ADS global technical regulation, translate China's technological advantages and industrial demands into international rules, and enhance China's discourse power in global autonomous driving governance.

4.2.5 Rationalize Central-Local Legislative Relations and Form a Coordinated Promotion Pattern

Clarify the division of powers between the central and local governments in autonomous driving governance: the central level focuses on bottom-line safety issues, formulates product access standards, liability allocation rules and data security red lines to provide superior law basis for local exploration; the local level, within the scope of central authorization, pioneers trials on issues such as operating area approval, commercial operation license issuance and infrastructure supporting, and elevates replicable and scalable experiences to national systems. This "central-local coordination" model not only avoids rule fragmentation caused by local fragmentation but also retains experimental space for institutional innovation.

5. Conclusion

The large-scale application of autonomous driving technology is driving profound changes in the global vehicle regulatory system. Comparative law investigations show that despite differences in national governance paths, preliminary consensus has been formed in core areas such as "multi-pillar safety assessment", "full lifecycle supervision" and "liability rule reconstruction". The adoption of the UN ADS global technical regulation in 2026 will mark a critical turning point in international rule coordination from "soft law guidelines" to "hard law constraints", shifting the focus of institutional competition from domestic legislation to the competition for international rule-making power.

China's autonomous driving governance is in a critical transition period from "local pioneering trials" to "national unified legislation". The current governance framework's problems, including the absence of superior laws, crude access standards, imperfect liability systems and insufficient international coordination, call for systematic institutional supply centered on the Autonomous Vehicles Law. The reconstructed governance system should resolve fragmentation dilemmas with a "unified framework", respond to differentiated demands with "classified regulation", and connect with the global rule system with "international coordination".

The core of autonomous driving governance has always been seeking a dynamic balance between safety and innovation. Legal systems should neither inhibit technological innovation through excessive prudence nor harm public safety through blind advancement. Only by absorbing foreign wisdom through comparative law, refining institutional consensus based on local practice, and shaping rule discourse from an international coordination perspective can we contribute Chinese solutions to transportation civilization in the autonomous driving era.

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